MFE5200 – Computational Methods in Financial Engineering
Summer Term, 2025-26

Assignment 2 (Solution)

Problem 1

From the given SDE of $S(t)$, we have

$$S(t_i) = S(0)\exp\left(\left(r - \frac{1}{2}\sigma^2\right)t_i + \sigma W(t_i)\right).$$

So,

$$\left(\prod_{i=1}^{n} S(t_i)\right)^{\frac{1}{n}} = S(0)\exp\left(\left(r - \frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n} + \frac{\sigma}{n}\sum_{i=1}^{n} W(t_i)\right).$$

Let X be $\begin{pmatrix} W(t_1) \\ W(t_2) \\ \vdots \\ W(t_n) \end{pmatrix}$.

So, $X \sim N(0, \mathbf{\Sigma})$, where

$$\mathbf{\Sigma} = \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_n \end{pmatrix}.$$

Let A be a $1 \times n$ matrix and $A = (1, 1, …., 1)$.

We have $\sum_{i=1}^{n} W(t_i) = AX$.

From the linear transformation property of the multivariate normal distribution, we have

$$\sum_{i=1}^{n} W(t_i) \sim N\left(0, A\mathbf{\Sigma}A^T\right) \text{ where } A\mathbf{\Sigma}A^T = \sum_{i=1}^{n}(2i-1)t_{n+1-i} = \sum_{i=1}^{n}(2n-2i+1)t_i.$$

Therefore,

$$\left(\prod_{i=1}^{n} S(t_i)\right)^{\frac{1}{n}} = \exp\left(\log S(0) + \left(r - \frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n} + \alpha Z\right),$$

where $Z \sim N(0, 1)$ and

$$\alpha^2 = \frac{\sigma^2}{n^2}\sum_{i=1}^{n}(2i-1)t_{n+1-i}.$$

Hence,

$$\left(\prod_{i=1}^{n} S(t_i)\right)^{\frac{1}{n}} \sim \mathrm{LN}\left(\log S(0) + \left(r - \frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n}, \alpha^2\right) \text{ and}$$

$$\begin{aligned} E\left[\left(\prod_{i=1}^{n} S(t_i)\right)^{\frac{1}{n}}\right] &= \exp\left(\log S(0) + \left(r - \frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n} + \frac{1}{2}\alpha^2\right) \\ &= S(0)\exp\left(\left(r - \frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n} + \frac{1}{2}\alpha^2\right). \end{aligned}$$

By using the given information, the pricing formula of the Asian option is given by

$$\begin{aligned} V &= e^{-rT}E^{Q}\left[\max\left(\bar{S}_G - K, 0\right)\right] \\ &= e^{-rT}\left[E^{Q}\left(\bar{S}_G\right)N(d_1) - KN(d_2)\right] \\ &= S(0)e^{-rT}\exp\left(\left(r - \frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n} + \frac{1}{2}\alpha^2\right)N(d_1) - Ke^{-rT}N(d_2), \end{aligned}$$

where

$$d_1 = \frac{\log\left(\frac{E^Q\left(\bar{S}_G\right)}{K}\right)+\frac{1}{2}\alpha^2}{\alpha} = \frac{\log\left(\frac{S(0)}{K}\right)+\left(r-\frac{1}{2}\sigma^2\right)\frac{\sum_{i=1}^{n} t_i}{n}+\alpha^2}{\alpha},$$

$$d_2 = d_1 - \alpha.$$

Problem 2

From the given dynamics of $S_1(t)$ and $S_2(t)$, we have

$$S_i(t) = S_i(0)\exp\left(\left(\mu_i - \frac{1}{2}\sigma_i^2\right)t + \sigma_i X_i(t)\right), \quad i = 1, 2.$$

$$\begin{aligned}
&\mathrm{Cov}\left(S_1(t), S_2(t)\right) \\
&= E\left(S_1(t)S_2(t)\right) - E\left(S_2(t)\right)E\left(S_2(t)\right) \\
&= E\left(S_1(0)S_2(0)\exp\left(\left(\mu_1+\mu_2-\frac{1}{2}\sigma_1^2-\frac{1}{2}\sigma_2^2\right)t+\sigma_1X_1(t)+\sigma_2X_2(t)\right)\right) - e^{\mu_1 t}S_1(0)e^{\mu_2 t}S_2(0) \\
&= S_1(0)S_2(0)\exp\left(\left(\mu_1+\mu_2-\frac{1}{2}\sigma_1^2-\frac{1}{2}\sigma_2^2\right)t\right)E\left(\exp\left(\sigma_1X_1(t)+\sigma_2X_2(t)\right)\right) \\
&-S_1(0)S_2(0)e^{(\mu_1+\mu_2)t}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\mathrm{Cov}\left(S_1(0.5), S_2(0.5)\right) \\
&= 30\exp\left(\left(0.08-\frac{1}{2}(0.3)^2-\frac{1}{2}(0.2)^2\right)(0.5)\right)E\left(\exp\left(0.3X_1(0.5)+0.2X_2(0.5)\right)\right) - 30e^{0.08\times 0.5} \\
&=30e^{0.0075}E\left(\exp\left(0.3X_1(0.5)+0.2X_2(0.5)\right)\right) - 30e^{0.04}.
\end{aligned}$$

Let $\mathbf{Y} = (Y_1, Y_2)^T$ be a 2-dimensional multivariate normal distribution with mean vector $\mathbf{0}$ and covariance matrix $\mathbf{\Sigma}$ where

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & -0.2 \\ -0.2 & 1 \end{pmatrix}.$$

So,

$$
\begin{aligned}
&E\left(\exp\left(0.3X_1(0.5)+0.2X_2(0.5)\right)\right)\\
&=E\left(\exp\left(0.3\sqrt{0.5}Y_1+0.2\sqrt{0.5}Y_2\right)\right)\\
&=E\left(\exp\left(\boldsymbol{\theta}^T\mathbf{Y}\right)\right), \quad \text{where } \boldsymbol{\theta}^T=\left(0.3\sqrt{0.5},0.2\sqrt{0.5}\right)\\
&=\exp\left(\frac{1}{2}\boldsymbol{\theta}^T\boldsymbol{\Sigma}\boldsymbol{\theta}\right)\\
&=\exp\left(\frac{1}{2}\left(0.3^2\times 0.5+0.2^2\times 0.5-2\times 0.2\times 0.3\times 0.2\times 0.5\right)\right)\\
&=e^{0.0265}.
\end{aligned}
$$

Hence, we have

$$
\begin{aligned}
&\mathrm{Cov}\left(S_1(0.5),S_2(0.5)\right)\\
&=30e^{0.0075}\times e^{0.0265}-30e^{0.04}\\
&=-0.1868.
\end{aligned}
$$

<u>Problem 3</u>

In the control variate method, we have

$$\overline{Y}(b)=\frac{1}{n}\sum_{i=1}^{n}\left(Y_i-b\left(X_i-E[X]\right)\right).$$

So,

$$\mathrm{Var}\left(\overline{Y}(b)\right)=\frac{\sigma_Y^2-2b\sigma_X\sigma_Y\rho_{XY}+b^2\sigma_X^2}{n}\equiv\sigma^2(b)..$$

The optimal value of b, b^*, is then given by

$$
\begin{aligned}
b^*&=\arg\min\sigma^2(b)\\
&=-\frac{-2\sigma_X\sigma_Y\rho_{XY}}{2\sigma_X^2}\\
&=\frac{\sigma_X\sigma_Y\rho_{XY}}{\sigma_X^2}\\
&=\frac{Cov(X,Y)}{Var(X)}.
\end{aligned}
$$

<u>Problem 4</u>

To show Y is the antithetic variable of X, you need to show that X and Y follow the same distribution and $Cov(X, Y) < 0$.

Consider

$$\begin{aligned} P(Y=0) &= P(X=1 \text{ and } Z=1) \\ &= p\left(\frac{1-p}{p}\right) \\ &= 1-p. \end{aligned}$$

$$\begin{aligned} P(Y=1) &= P(X=1 \text{ and } Z=0) + P(X=0 \text{ and } Z=1) + P(X=0 \text{ and } Z=0) \\ &= p\left(\frac{2p-1}{p}\right) + (1-p)\left(\frac{1-p}{p}\right) + (1-p)\left(\frac{2p-1}{p}\right) \\ &= p. \end{aligned}$$

Therefore, X and Y follow the same distribution.

$$\begin{aligned} Cov(X,Y) &= E(XY) - E(X)E(Y) \\ &= E(X(1-XZ)) - E(X)E(1-XZ) \\ &= E(X) - E(X^2Z) - E(X) + E(X)E(XZ) \\ &= E(X) - E(X^2)E(Z) - E(X) + (E(X))^2 E(Z) \\ &= E(Z)\left((E(X))^2 - E(X^2)\right) \\ &= -E(Z)Var(X) \\ &= -\left(\frac{1-p}{p}\right)p(1-p) \\ &= -(1-p)^2 < 0. \end{aligned}$$

Hence, Y is the antithetic variable of X.

Problem 5

(a)

Let $Y_i = \mathbf{1}_{\{Z_i \geq 0.8\}}$.

Since the function $f(x) = \mathbf{1}_{\{x \geq 0.8\}}$ is a monotonic function, so the antithetic variable is $\tilde{Y}_i = 1_{\{Z_i \leq -0.8\}}$.

The antithetic variates estimator is given by

$$\hat{Y}_{AV} = \frac{1}{n}\sum_{i=1}^{n}\frac{Y_i + \tilde{Y}_i}{2}.$$

(b)

$$\begin{aligned}
\mathrm{Var}\left(\hat{Y}_{AV}\right) &= \frac{1}{4n^2}\sum_{i=1}^{n}\mathrm{Var}\left(Y_i + \tilde{Y}_i\right) \\
&= \frac{1}{4n}\left(2\mathrm{Var}\left(Y_1\right) + 2\mathrm{Cov}\left(Y_1, \tilde{Y}_1\right)\right) \\
&= \frac{1}{2n}\left(\mathrm{Var}\left(Y_1\right) + \mathrm{Cov}\left(Y_1, \tilde{Y}_1\right)\right).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\mathrm{Var}\left(\overline{Y}\right) &= \frac{1}{4n^2}\sum_{i=1}^{2n}\mathrm{Var}\left(Y_i\right) \\
&= \frac{1}{4n^2}\left(2n\mathrm{Var}\left(Y_1\right)\right) \\
&= \frac{1}{2n}\mathrm{Var}\left(Y_1\right) \\
&= \frac{1}{2n}\left(E\left[Y_1^2\right] - \left(E\left[Y_1\right]\right)^2\right) \\
&= \frac{1}{2n}\left(P\left(Z_1 \geq 0.8\right) - \left(P\left(Z_1 \geq 0.8\right)\right)^2\right) \\
&= \frac{1}{2n}\left(\left(1 - 0.7881\right) - \left(1 - 0.7881\right)^2\right) \\
&= \frac{0.0835}{n}.
\end{aligned}$$

The magnitude of variance change is

$$\operatorname{Var}\left(\hat{Y}_{AV}\right) - \operatorname{Var}\left(\overline{Y}\right) = \frac{1}{2n}\operatorname{Cov}\left(Y_1, \tilde{Y}_1\right).$$

We have

$$\begin{aligned}
\operatorname{Cov}\left(Y_1, \tilde{Y}_1\right) &= E\left[Y_1\tilde{Y}_1\right] - E\left[Y_1\right]E\left[\tilde{Y}_1\right] \\
&= E\left[\mathbf{1}_{\{Z_1 \geq 0.8\}}\mathbf{1}_{\{Z_1 \leq -0.8\}}\right] - E\left[\mathbf{1}_{\{Z_1 \geq 0.8\}}\right]E\left[\mathbf{1}_{\{Z_1 \leq -0.8\}}\right] \\
&= 0 - P\left(Z_1 \geq 0.8\right)P\left(Z_1 \leq -0.8\right) \\
&= -\left(0.2119\right)\left(0.2119\right) \\
&= -0.0449.
\end{aligned}$$

So,

$$\begin{aligned}
\operatorname{Var}\left(\hat{Y}_{AV}\right) - \operatorname{Var}\left(\overline{Y}\right) &= \frac{-0.0449}{2n} \\
&= -\frac{0.0225}{n}.
\end{aligned}$$

The percentage of the variance change in the antithetic variates method is

$$\begin{aligned}
\text{\% of the variance change} &= \frac{-\frac{0.0225}{n}}{\frac{0.0835}{n}} \times 100\% \\
&= -26.9461\%.
\end{aligned}$$

Therefore, the percentage of the variance reduction in the antithetic variates method is 29.9461%.

Problem 6

(a)

Let $Y = e^U$.

Partition the unit interval [0, 1] into 2 strata

$$A_1 = \left[0, \frac{1}{2}\right], \quad A_2 = \left(\frac{1}{2}, 1\right].$$

With the proportional allocation, we have

$$p_i = \frac{1}{2} \quad \text{and} \quad n_i = p_i(2n) = n, \quad i = 1, 2.$$

The stratified estimator is given by

$$\hat{Y} = \frac{1}{2n} \sum_{i=1}^{2} \sum_{j=1}^{n} Y_{ij}$$

where

$$Y_{1j} = e^{V_{1j}} \text{ and } Y_{2j} = e^{V_{2j}}.$$

(b)

To evaluate the magnitude of the variance reduction, we need

$$\mu_1 = E\left[Y_{1j}\right] = E\left[e^{V_{1j}}\right] = \int_0^{0.5} 2e^x dx = 2\left(e^{\frac{1}{2}} - 1\right).$$

$$\mu_2 = E\left[Y_{2j}\right] = E\left[e^{V_{2j}}\right] = \int_{0.5}^{1} 2e^x dx = 2\left(e - e^{\frac{1}{2}}\right).$$

So, the magnitude of the variance reduction is

$$\frac{1}{2n}\left[\sum_{i=1}^{2} p_i \mu_i^2 - \left(\sum_{i=1}^{2} p_i \mu_i\right)^2\right]$$

$$= \frac{1}{2n}\left[\frac{1}{2}\left(4\left(e^{\frac{1}{2}} - 1\right)^2 + 4\left(e - e^{\frac{1}{2}}\right)^2\right) - \frac{1}{4}\left(2\left(e^{\frac{1}{2}} - 1\right) + 2\left(e - e^{\frac{1}{2}}\right)\right)^2\right]$$

$$= \frac{0.0886}{n}.$$

(c)
The optimal allocation is given by

$$q_i^* = \frac{p_i \sigma_i}{\sum_{l=1}^{2} p_l \sigma_l}$$

where σ_i^2= Var(Y_{ij}).

$$\begin{aligned}
Var\left(Y_{1j}\right) &= Var\left(e^{V_{1j}}\right) \\
&= E\left[e^{2V_{1j}}\right] - \left(E\left[e^{V_{1j}}\right]\right)^2 \\
&= \int_0^{0.5} 2e^{2x}dx - 4\left(e^{\frac{1}{2}} - 1\right)^2 \\
&= \left(e-1\right) - 4\left(e^{\frac{1}{2}} - 1\right)^2 \\
&= 0.0349.
\end{aligned}$$

$$\begin{aligned}
Var\left(Y_{2j}\right) &= Var\left(e^{V_{2j}}\right) \\
&= E\left[e^{2V_{2j}}\right] - \left(E\left[e^{V_{2j}}\right]\right)^2 \\
&= \int_{0.5}^{1} 2e^{2x}dx - 4\left(e - e^{\frac{1}{2}}\right)^2 \\
&= \left(e^2 - e\right) - 4\left(e - e^{\frac{1}{2}}\right)^2 \\
&= 0.0949.
\end{aligned}$$

So,

$$q_1^* = \frac{0.5\sqrt{0.0349}}{0.5\left(\sqrt{0.0349} + \sqrt{0.0949}\right)} = 0.3775, \quad \text{and}$$

$$q_2^* = \frac{0.5\left(\sqrt{0.0949}\right)}{0.5\left(\sqrt{0.0349} + \sqrt{0.0949}\right)} = 0.6225.$$

The stratified estimator corresponding to the optimal allocation is given by

$$\hat{Y}^* = \frac{1}{4n}\left(\frac{1}{q_1^*}\sum_{j=1}^{2nq_1^*} Y_{1j} + \frac{1}{q_2^*}\sum_{j=1}^{2nq_2^*} Y_{2j}\right).$$

Problem 7

Rewrite

$$\Pr\left(X \ge m\alpha\right) = E\left(1_{\{X \ge m\alpha\}}\right).$$

Since the binomial distribution with parameters $\left(m, \overline{p}\right)$ is taken as the alternative distribution, the importance sampling estimate is

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n} 1_{\{Y_i \ge m\alpha\}} \frac{\dfrac{m!}{\left(m-Y_i\right)!Y_i!} p^{Y_i}\left(1-p\right)^{m-Y_i}}{\dfrac{m!}{\left(m-Y_i\right)!Y_i!} \overline{p}^{Y_i}\left(1-\overline{p}\right)^{m-Y_i}} \\
&= \frac{1}{n}\sum_{i=1}^{n} 1_{\{Y_i \ge m\alpha\}} \frac{p^{Y_i}\left(1-p\right)^{m-Y_i}}{\overline{p}^{Y_i}\left(1-\overline{p}\right)^{m-Y_i}} \\
&= \frac{1}{n}\sum_{i=1}^{n} 1_{\{Y_i \ge m\alpha\}} \left(\frac{p}{\overline{p}}\right)^{Y_i}\left(\frac{1-p}{1-\overline{p}}\right)^{m-Y_i}.
\end{aligned}$$

Problem 8

(a)

We have

$$1_{\{x \ge b\}} f\left(x\right) = \begin{cases} 0 & x < b \\ f\left(x\right) & x \ge b \end{cases}.$$

Since $f(x)$ is a nonnegative monotonic decreasing function over $[0, \infty]$, $x^* = b$.

(b)

Let $g(x)$ be the pdf of $N(b, 1)$.

$$g\left(x\right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(x-b\right)^2}{2}}.$$

With the alternative pdf $g(x)$, the importance sampling estimator is given by

$$
\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{1}_{\{X_i \ge b\}} f(X_i)}{g(X_i)} \\
&= \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}_{\{X_i \ge b\}} \exp\left(-\frac{X_i^{\,2}}{2} + \frac{(X_i - b)^2}{2}\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}_{\{X_i \ge b\}} \exp\left(-bX_i + \frac{b^2}{2}\right).
\end{aligned}
$$

Problem 9

(a)

If X is Bernoulli with parameter p, then the probability mass function (pmf) of X is given by

$$
p(x) = P(X = x) = \begin{cases} 1-p & \text{if } x = 0; \\ p & \text{if } x = 1; \\ 0 & \text{otherwise.} \end{cases}
$$

The exponential tilt family of the random variable X is

$$
\begin{aligned}
p_\theta(x) &= \frac{1}{E\left[e^{\theta X}\right]} e^{\theta x} p(x) \\
&= \frac{1}{pe^{\theta} + (1-p)} e^{\theta x} p(x).
\end{aligned}
$$

So,

$$
p_\theta(x) = \begin{cases} \dfrac{1-p}{pe^{\theta} + (1-p)} & \text{if } x = 0; \\ \dfrac{pe^{\theta}}{pe^{\theta} + (1-p)} & \text{if } x = 1; \\ 0 & \text{otherwise.} \end{cases}
$$

(b)

The exponential tilt family of the random variable X is

$$\begin{aligned} f_\theta(x) &= \frac{1}{E\left[e^{\theta X}\right]} e^{\theta x} f(x) \\ &= \frac{1}{\frac{\lambda}{(\lambda-\theta)}} e^{\theta x} \lambda e^{-\lambda x} \\ &= (\lambda-\theta) e^{-(\lambda-\theta)x}. \end{aligned}$$

<u>Problem 10</u>

$\hat{\theta}$ is the solution of θ in the following equation,

$$0 = \frac{1}{N}\sum_{k=1}^{N} h(X_k) \frac{\partial}{\partial\theta} \log f_\theta(X_k). \qquad (1)$$

Since $\{f_\theta\}$ is an exponential tilt family, we have

$$f_\theta(x) = \frac{1}{E\left[e^{\theta X}\right]} e^{\theta x} f(x).$$

So,

$$\log f_\theta(X_k) = \log f(x) + \theta x - H(\theta).$$

Eq. (1) then becomes

$$\begin{aligned} 0 &= \frac{1}{N}\sum_{k=1}^{N} h(X_k) \frac{\partial}{\partial\theta}\left(\log f(X_k) + \theta X_k - H(\theta)\right) \\ 0 &= \frac{1}{N}\sum_{k=1}^{N} h(X_k)\left(X_k - H'(\theta)\right) \\ H'(\theta) &= \frac{\sum_{k=1}^{N} h(X_k) X_k}{\sum_{k=1}^{N} h(X_k)}. \end{aligned}$$

From Eq. (7.12), we have

$$0 = \frac{1}{N}\sum_{k=1}^{N} h\left(Y_k\right)\ell_{\hat{\theta}^j}\left(Y_k\right)\left(\frac{\partial}{\partial\theta}\left(\log f\left(Y_k\right)+\theta Y_k - H\left(\theta\right)\right)\right)\Bigg|_{\theta=\hat{\theta}^{j+1}}$$

$$0 = \frac{1}{N}\sum_{k=1}^{N} h\left(Y_k\right)\ell_{\hat{\theta}^j}\left(Y_k\right)\left(Y_k - H'\left(\hat{\theta}^{j+1}\right)\right)$$

$$H'\left(\hat{\theta}^{j+1}\right) = \frac{\sum_{k=1}^{N} h\left(Y_k\right)\ell_{\hat{\theta}^j}\left(Y_k\right)Y_k}{\sum_{k=1}^{N} h\left(Y_k\right)\ell_{\hat{\theta}^j}\left(Y_k\right)}.$$